

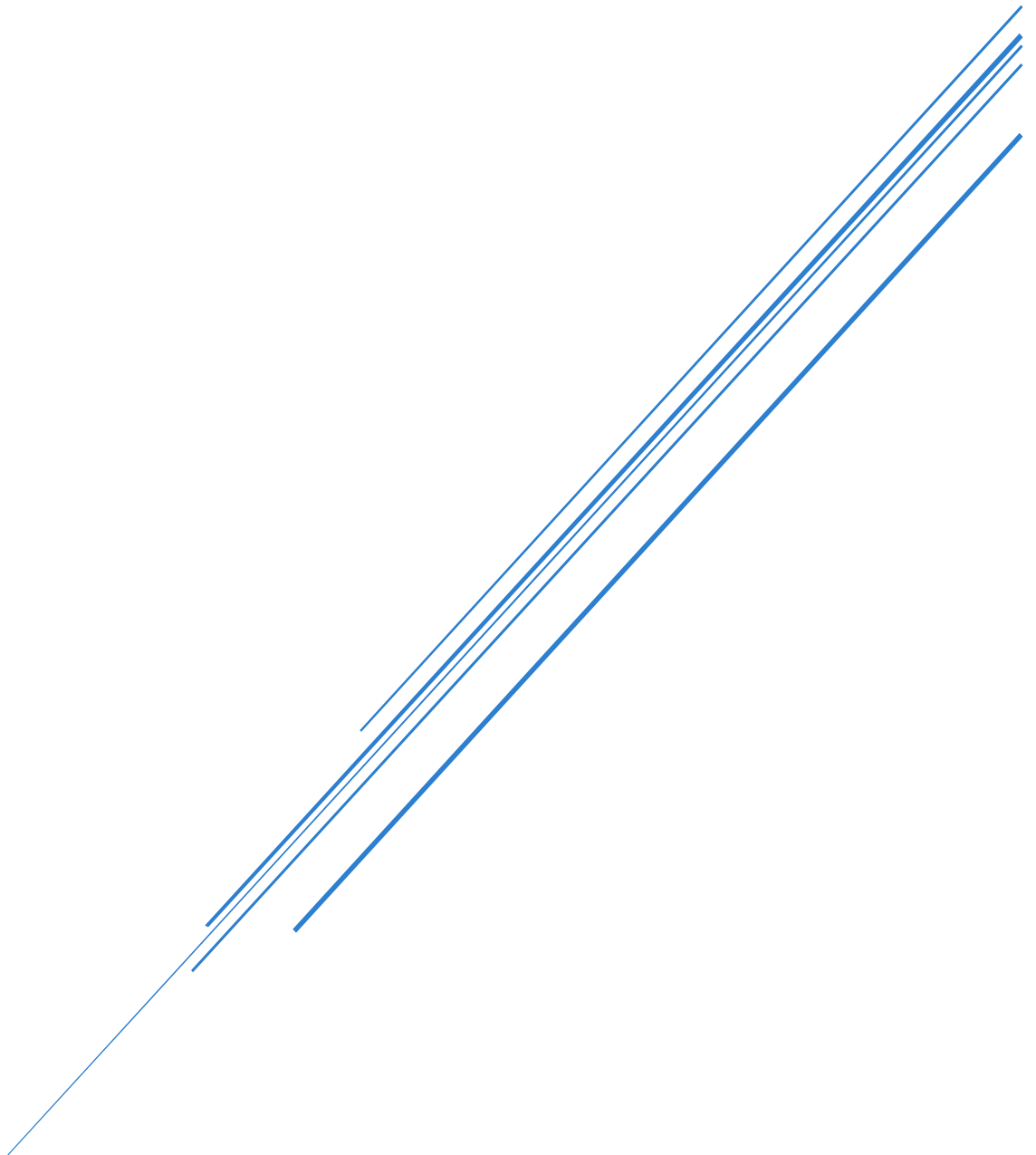
INFORMATION SYSTEMS: ICE

TASK 6

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Predictive Approach to System Development

The predictive approach to system development is a traditional method where the project is planned in detail before development begins. It follows a structured sequence of phases such as planning, analysis, design, implementation, testing, and maintenance. This approach is commonly associated with the Waterfall model because each phase is completed before moving to the next phase.

1. Key Characteristics

One of the main characteristics of the predictive approach is extensive planning and documentation. Project requirements are identified at the beginning, and the development team follows a fixed schedule and budget throughout the project. The process is highly structured, with clear objectives, timelines, and deliverables. Progress is measured against the original plan, making project control easier for managers.

Another important characteristic is limited change during development. Since requirements are defined early, modifications later in the project are often difficult and costly. The predictive approach therefore works best when project requirements are stable and well understood from the start.

2. Strengths of the Predictive Approach

A major advantage of the predictive approach is strong project control. Because the project follows a detailed plan, managers can estimate costs, timelines, and resource requirements more accurately. This makes it suitable for large organisations that require strict monitoring and reporting.

The approach also provides comprehensive documentation. Detailed records of requirements, designs, and testing processes make maintenance and future upgrades easier. Documentation is especially important in industries such as banking, healthcare, and government, where compliance and accountability are essential.

In addition, risks can be identified early during the planning stage. Since the entire project is analysed before development starts, potential issues can often be anticipated and managed in advance.

Weaknesses of the Predictive Approach

Despite its advantages, the predictive approach has several limitations. The biggest weakness is its lack of flexibility. Changes in customer requirements are difficult to accommodate once development has started. This can lead to delays and increased costs if adjustments are needed later.

Another weakness is limited client involvement during development. Clients are usually involved mainly during the requirements phase and final delivery, which may result in a product that does not fully meet changing user needs.

The predictive approach can also be time-consuming because of the heavy emphasis on planning and documentation before actual development begins. If requirements are unclear at the start, the final system may not satisfy user expectations.

3. Application Scenario

The predictive approach is most suitable for projects with clearly defined and stable requirements. For example, a government payroll management system would benefit from this approach because the regulations, procedures, and security requirements are usually fixed and require detailed documentation and strict project control.

Adaptive Approach To System Development

The adaptive approach to system development is a modern, flexible method that focusses on rapid delivery and continuous improvement. Unlike the predictive approach it assumes that the requirements will evolve as the project progresses. It is frequently associated with agile methodologies, where development is broken down into small, manageable increments or iterations.

Characteristics

One of the main characteristics of the adaptive model is its cyclicality and incrementality. Instead of building the entire system all at once, the system is broken down into shorter cycles, usually taking two to four weeks each. At the end of each cycle, a working piece of the system is created and evaluated.

High collaboration is another characteristic that stands out in this model. The development team collaborates closely with the client and other stakeholders during the whole process. The requirements are not predefined at the beginning; rather, they are kept in an always-changing list referred to as a backlog.

Adaptive Approach Strengths

The biggest advantage of the adaptive method is its great flexibility. As the development is conducted in small cycles, there is plenty of room for manoeuvring, and any new requirements and changes in market demands can be easily incorporated into the project.

Moreover, satisfaction among the clients is much higher thanks to the involvement of stakeholders during the development process. With regular access to pieces of the software under development, clients can provide input and feedback to improve the result and avoid failure.

Furthermore, risk management is improved through early and continuous testing. Because issues are identified and get fixed within it each iteration, the likelihood of a major system failure at the end of the project is greatly reduced compared to more rigid methodologies.

Weaknesses of the Adaptive Approach

One weakness of the adaptive approach is the potential for “scope creep” due to the requirements constantly changing and there is no fixed end-date for specific features, projects can sometimes also grow beyond their original intent, ultimately leading to increased costs and timeframes if not managed strictly.

This approach also tends to produce less formal documentation due to the focus on “working software over comprehensive documentation”, further maintenance or transitions to new team members can be more challenging if the code were to be not well-documented or technical specifications are sparse.

Lastly, the adaptive approach typically requires a high level of commitment and time from the client. For organisations where stakeholders are typically busy and cannot meet with the development team regularly, this approach could potentially stall or lose its effectiveness.

Application Scenario

The adaptive approach is most suitable for projects in fast-paced or innovative environments where requirements are likely to change. For example, the development of a new mobile social media platform/application would benefit from this approach. It allows for the developers to launch a basic version, quickly add or change features based on what is most popular in the market and observe user behaviour